

Week 3: Environment Setup & Project Initialization

Objective

Set up the Java development environment, configure Java Swing, and initialize the Scientific Calculator project for further development.

Workflow

1.

a) Install JDK

- Install the Java Development Kit (JDK).
- Verify Java installation from the terminal:

```
java -version  
javac -version
```

b) Install and Configure IDE

- Install a Java-supported IDE such as IntelliJ IDEA, Eclipse, or NetBeans.
- Create a new Java project.
- Select the installed JDK as the project SDK.

2. Configure Java Swing

- Use Java Swing for the graphical user interface.
- No separate Swing library is required because Swing is included with the JDK.
- Import the required Swing, AWT, and event packages.

3.

- **Create Project Structure**

Suggested structure:

```
ScientificCalculator/  
├── src/  
│   └── Scientific_calculator.java  
└── README.md
```

- **Create Main Class**

- Create the main class: `Scientific_calculator`
- Extend `JFrame` to create the calculator window.
- Initialize the application from the `main()` method.
- Launch the GUI using `SwingUtilities.invokeLater()`.

Week 3 Deliverables

- JDK installed and verified
- Java IDE configured
- Swing environment configured
- Project structure created
- Main Java class initialized